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Current Sensing Circuit With Temperature Coefficient Compensation

Abstract

A current sensing circuit includes a first field-effect transistor (FET) coupled to a first current source, an inductor, and a power FET, and includes a second FET coupled to a second current source. The current sensing circuit also includes a temperature circuit coupled to the first FET and the second FET. The temperature circuit is configured to compensate for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and the power FET.

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Background/Summary

BACKGROUND

Technical Field

[0001] Embodiments described herein relate to integrated circuits, and more particularly, to techniques for measuring or sensing currents.

Description of the Related Art

[0002] Modern computer systems may include multiple circuit blocks designed to perform various functions. For example, such circuit blocks may include processors, processor cores configured to execute software or program instructions. Additionally, the circuit blocks may include memory circuits, mixed-signal or analog circuits, and the like.

[0003] In some computer systems, the circuit blocks may be designed to operate at different power supply voltage levels. Power management circuits may be included in such computer systems to generate and monitor varying power supply voltage levels for the different circuit blocks.

[0004] Power management circuits often include one or more power converter circuits configured to generate regulator voltage levels on respective power supply signals using a voltage level of an input power supply signal. Such regulator circuits may employ multiple passive circuit elements, such as inductors, capacitors, and the like.

SUMMARY OF VARIOUS EMBODIMENTS

[0005] Various embodiments of a current sensing circuit are disclosed. Broadly speaking, a current sensing circuit and/or a temperature circuit for the current sensing circuit are contemplated. The current measuring circuit may use the temperature circuit to compensate for differences in temperature coefficients between one or more field-effect transistors (FETs) of the current measuring circuit. The current measuring circuit may include a first FET, a second FET, an inductor, and the temperature circuit. The temperature circuit may include a first set of FETs, a second set of FETs, a first resistor, and a second resistor.

[0006] Various embodiments of a power converter circuit are also disclosed. Broadly speaking, the power converter circuit for sourcing a charge current to another device/circuit are contemplated. The power converter circuit may include a current sensing circuit configured to measure, detect, sense, etc., the amount of charge current sourced to the other device/circuit. The current measuring circuit includes a temperature circuit to compensate for differences in temperature coefficients between one or more field-effect transistors (FETs) of the current measuring circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The following detailed description makes reference to the accompanying drawings, which are now briefly described.

[0008] FIG. 1 illustrates a diagram of an example power converter circuit, in accordance with one or more embodiments of the present disclosure.

[0009] FIG. 2 illustrates a diagram of an example voltage regulator circuit, in accordance with one or more embodiments of the present disclosure.

[0010] FIG. 3 illustrates a block diagram of an example control circuit for a power converter circuit, in accordance with one or more embodiments of the present disclosure.

[0011] FIG. 4 illustrates a block diagram of an example current sensing circuit, in accordance with one or more embodiments of the present disclosure.

[0012] FIG. 5 illustrates a block diagram of an example current sensing circuit, in accordance with one or more embodiments of the present disclosure.

[0013] FIG. 6 illustrates a block diagram of an example temperature circuit, in accordance with one or more embodiments of the present disclosure.

[0014] FIG. 7 illustrates a flow diagram depicting an embodiment of a method for sensing a current, in accordance with one or more embodiments of the present disclosure.

[0015] FIG. 8 illustrates a block diagram of an example computer system, in accordance with one or more embodiments of the present disclosure.

[0016] While the disclosure is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the disclosure to the particular form illustrated, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to.

[0017] Various units, circuits, or other components may be described as “configured to” perform a task or tasks. In such contexts, “configured to” is a broad recitation of structure generally meaning “having circuitry that” performs the task or tasks during operation. As such, the unit/circuit/component can be configured to perform the task even when the unit/circuit/component is not currently on. In general, the circuitry that forms the structure corresponding to “configured to” may include hardware circuits. Similarly, various units/circuits/components may be described as performing a task or tasks, for convenience in the description. Such descriptions should be interpreted as including the phrase “configured to.” Reciting a unit/circuit/component that is configured to perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112, paragraph (f) interpretation for that unit/circuit/component. More generally, the recitation of any element is expressly intended not to invoke 35 U.S.C. § 112, paragraph (f) interpretation for that element unless the language “means for” or “step for” is specifically recited.

[0018] As used herein, the term “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect the determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor that is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. The phrase “based on” is thus synonymous with the phrase “based at least in part on.”

DETAILED DESCRIPTION OF EMBODIMENTS

[0019] Computer systems may include multiple circuit blocks configured to perform specific functions. Such circuit blocks may be fabricated on a common substrate and may employ different power supply voltage levels. Power management units (commonly referred to as “PMUs”) may include multiple power converter circuits configured to generate regulated voltage levels for various power supply signals. Such power converter circuits may employ regulator circuit that includes both passive circuit elements (e.g., inductors, capacitors, etc.) as well as active circuit elements (e.g., transistors, diodes, etc.).

[0020] Different types of voltage regulator circuits may be employed based on power requirements of load circuits, available circuit area, and the like. One type of commonly used voltage regulator circuit is a buck converter circuit. Such buck converter circuits include multiple devices and a switch node that is coupled to a regulated power supply node via an inductor and/or other circuit used to sense/measure current. Particular ones of the multiple devices are then activated to

periodically charge and discharge the switch node in order to maintain a desired voltage level on power supply node.

[0021] To determine the duration of either the charge cycle or discharge cycle, current mode control may be used in some power converter circuits in order to provide a desired transient response of the power converter circuit as well as balance currents in multi-phase power converter circuits. In a power converter circuit using current mode control, control circuits may generate a control current whose values is based, at least in part, on a comparison of a voltage level of the regulated power supply node and a reference voltage. The control current may then be compared to a current that is flowing to an output of the power converter circuit to determine the duration of the charge or discharge cycle.

[0022] Because the power converter circuit uses the current flowing to the output of the to determine the duration of the charge cycle (e.g., when to start a charge cycle) or discharge cycle (e.g., when to halt a charge cycle), it may be important to accurately sense (e.g., detect, measure, etc.,) the amount of current flowing to the output of the power converter circuit. Sensing the amount of current flowing to the output of the power converter circuit more accurately may allow the power converter circuit to operate more efficiently, use less power, etc.

[0023] Different circuits will often have different temperature coefficients. A temperature coefficient may indicate how a property of a circuit changes when the temperature of the circuit changes. For example, different field-effect transistors (FETs) will have different temperature coefficients. The temperature coefficient of a FET may indicate the change in a resistance of the FET when the temperature of the FET changes.

[0024] As discussed above, the power converter circuit may provide current to another device/circuit via one or more FETs (e.g., via a power FET). As current flows through the FET, the temperature of the FET may increase which may change the resistance of the FET. A current sensing circuit may use one or more FETs to sense (e.g., e.g., one or more sense FETs) the amount of current provided to the other device via the power FET. The resistance of sense FETs may also change as their temperatures increase. Because the sense FETs and the power FET may have different temperature coefficients, this may affect the accuracy of the current sensing circuit.

[0025] The embodiments illustrated in the drawings and described below may provide techniques for operating a current sensing circuit that has temperature coefficient compensation. For example, the current sensing circuit may compensate for the differences in the temperature coefficients of the FETs (e.g., of the sense FETs and the power FET). By compensating for the differences in the temperature coefficients, accuracy of the current sensing circuit may be improved.

[0026] FIG. 1 illustrates a block diagram of an example power converter circuit, in accordance with one or more embodiments of the present disclosure. As illustrated, power converter circuit **100** includes control circuit **101**, voltage regulator circuit **102**, and current sensing circuit **120**.

[0027] Voltage regulator circuit **102** includes switch node **105** coupled to regulated power supply node **110** via current sensing circuit **120**. In various embodiments, voltage regulator circuit **102** is configured, in response to an initiation of charge cycle **106**, to source charge current **103** to switch node **105**. It is noted that although a single voltage regulator circuit is depicted in the embodiment of FIG. 1, in other embodiments, multiple voltage regulator circuits (collectively “phase units” or “phase circuits”) may be coupled to regulated power supply node **110**, in parallel, and operated with different timings (or “phases”).

[0028] As noted above, the duration of charge and discharge cycles in a power converter circuit may be determined using current control. As illustrated in FIG. 1, control circuit **101** is configured, in response to an initiation of charge cycle **106**, to generate control current **111** using a voltage level of regulated power supply node **110** and reference voltage level **109**. In various embodiments, control circuit **101** initiates charge cycle **106** in response to an assertion of a clock or other signal (e.g., a timing signal).

[0029] Control circuit **101** is also configured to generate compensation current **108**. In various

embodiments, compensation current **108** may be used to perform slope compensation on sensed inductor current **112**. Compensation current **108** may be an increasing ramp signal with a fixed slope that is activated in response to the assertion of the clock or other timing signal. As used herein, sensed inductor current **112** is a current that is flowing through a current sensing circuit **120** during a charge cycle. In various embodiments, sensed inductor current **112** may be inferred from a voltage level of switch node **105**, measured using a current sensing circuit **120**. The current sensing circuit **120** may measure the voltage level of the switch node **105** using a voltage drop across a resistor in series with an inductor, or any other suitable circuit or technique for sensing and/or measuring current.

[0030] Current sensing circuit **120** is configured to sense (e.g., detect, measure, estimate, etc.) the current that is generated by the power converter circuit **100**. The power converter circuit **100** (e.g., the control circuit **101**) may control the operation of the power converter circuit **100** based on the measured current. For example, the duration of charge and discharge cycles in a power converter circuit **100** may be determined based on the current that is sensed by the current sensing circuit.

[0031] Although the current sensing circuit **120** is illustrated as separate from the control circuit **101** and the voltage regulator circuit **102**, the current sensing circuit **120** may be part of the control circuit **101** and/or the voltage regulator circuit **102**. For example, the current sensing circuit **120** may be part of the control circuit **101**. In another example, the current sensing circuit **120** may be part of the voltage regulator circuit **102**. In a further example, a portion of the current sensing circuit **120** may be included in the control circuit **101** and another portion of the current sensing circuit **120** may be included in the voltage regulator circuit **102**. The current sensing circuit **120** is discussed in more detail below.

[0032] In addition to generating compensation current **108**, control circuit **101** is also configured to halt and/or start charge cycle **106** using control current **111**, sensed inductor current **112**, and compensation current **108**. In various embodiments, control circuit may combine sensed inductor current **112** and compensation current **108**, and compare a result current to control current **111**.

[0033] As discussed above, the power converter circuit **100** uses the current flowing to the regulated power supply node **110** (e.g., and output of the power converter circuit **100**) to determine the duration of the charge cycle **106**. Sensing the amount of current flowing to the output of the power converter circuit more accurately may allow the power converter circuit **100** to operate more efficiently, use/waste less power, to generate less heat, etc.

[0034] Because different FETs have different temperature coefficients (e.g., have different resistances at a particular temperature), it may be more difficult to sense the amount of current flowing to the regulated power supply node **110**. In one embodiment, the current sensing circuit may compensate for the differences in the temperature coefficients of various FETs. By compensating for the differences in the temperature coefficients, accuracy of the operation of the power converter circuit **100** may be improved (e.g., the power converter circuit **100** may operate more efficiently).

[0035] FIG. 2 illustrates a diagram of an example voltage regulator circuit **102**, in accordance with one or more embodiments of the present disclosure. Voltage regulator circuits, such as voltage regulator circuit **102**, may be designed according to one of various design styles. A schematic diagram of a particular embodiment of voltage regulator circuit **102** is depicted in FIG. 2. As illustrated, voltage regulator circuit **102** includes devices **201** and **202**, which are both coupled to switch node **105**, and controlled by control signal **205**.

[0036] In various embodiments, control circuit **101** may generate control signal **205**, which is used to activate one of devices **201** and **202** during charge and discharge cycles. During a charge cycle, current is sourced from input power supply node **203** to regulated power supply node **110**, and during a discharge cycle, current is sunk from regulated power supply node **110** into ground supply node **204**. Alternating between charge and discharge cycles, and adjusting the duration of either of the charge or discharge cycles may maintain a desired voltage level maintained on regulated power

supply node **110**.

[0037] Device **201** is coupled between input power supply node **203** and switch node **105**, and is controlled by control signal **205**. During a charge cycle, control signal **205** is asserted, which activates device **201** and couples input power supply node **203** to switch node **105**, thereby charging switch node **105** by allowing a current to flow from input power supply node **203** to switch node **105**, and then onto regulated power supply node **110**. As described below in more detail, the duration of the charge cycle may be based on a comparison of a generated current to a combination of generated and sensed currents.

[0038] In one embodiment, the switch node **105** may be coupled to current sensing circuit **120**. The current sensing circuit **120** may measure, sense, or detect the current flowing through the regulated power supply node **110**, as discussed in more detail below.

[0039] As used herein, asserting, or an assertion of, a signal refers to setting the signal to a particular voltage level that activates a circuit or device coupled to the signal. The particular voltage level may be any suitable value. For example, in the case where device **201** is p-channel MOSFET, control signal **205** may be set to a voltage at or near ground potential when activated.

[0040] Device **202** is coupled between switch node **105** and ground supply node **204**, and is also controlled by control signal **205**. During a discharge cycle, control signal **205** is set to a voltage level, which activates device **202** and couples switch node **105** to ground supply node **204**, thereby providing a conduction path from regulated power supply node **110** into ground supply node **204**. While device **202** is active, current flows from regulated power supply node **110** into ground supply node **204**, decreasing the voltage level of regulated power supply node **110**.

[0041] Device **201** and device **202** may be particular embodiments of MOSFETs. In particular, device **201** may be a particular embodiment of a p-channel MOSFET and device **202** may be a particular embodiment of an n-channel MOSFET. Although only two devices are depicted in the embodiment of FIG. 2, in other embodiments, any suitable number of devices, coupled in series or parallel, may be employed to achieve particular electrical characteristics (e.g., on-resistance of the devices).

[0042] FIG. 3 illustrates a block diagram of an example control circuit **101** for a power converter circuit, in accordance with one or more embodiments of the present disclosure. As illustrated, control circuit **101** includes latch circuit **301**, comparator circuit **302**, comparator circuit **303**, and compensation circuit **304**.

[0043] Latch circuit **301** may be a particular embodiment of a Set-Reset (SR) latch configured to set control signal **205** to a low logic value in response to an assertion of clock signal **306**.

Additionally, Latch circuit **301** is configured to set control signal **205** to a high logic level in response as assertion of reset signal **310** on node **308**.

[0044] Latch circuit **301** may be designed according to one of various design styles. In various embodiments, latch circuit **301** may include multiple logic gates, such as, cross-coupled NAND gates, or any other suitable combination of logic gates and/or MOSFETs to implement the functionality described above.

[0045] Comparator circuit **302** is coupled to latch circuit **301** via node **308**, and may be a particular embodiment of a differential amplifier configured to generate reset signal **310** on node **308** using control current **111** and a combination of compensation current **108**, and sensed inductor current **112**. In various embodiments, comparator circuit **302** may be configured to set reset signal **310** to a particular digital voltage level using results of comparing control current **111** to the combination of compensation current **108**, and sensed inductor current **112**. For example, when a value of control current **111** is substantially the same as the combination of compensation current **108**, and sensed inductor current **112**, comparator circuit **302** may set the voltage level of reset signal **310** to a voltage level corresponding to a high logic level.

[0046] Comparator circuit **303** is coupled to comparator circuit **302** via node **309**, and may be a particular embodiment of a transconductance amplifier configured to generate control current **111**

in node **309**. The value of control current **111** may be based, at least in part, on a comparison of reference voltage level **109** and the voltage level of regulated power supply node **110**. In various embodiments, comparator circuit **303** may amplify a difference between reference voltage level **109** and the voltage level of regulated power supply node **110**, and convert the difference in voltage levels to control current **111**.

[0047] Compensation circuit **304** is coupled to clock signal **306** and switch node **105**, and is configured to generate compensation current **108**. In various embodiments, compensation circuit **304** is configured to generate compensation current **108** in response to an assertion of clock signal **306**. Compensation circuit **304** may be further configured to source (or add) compensation current **108** to sensed inductor current **112** to generate a sum of the two currents. Switch node **105** is also coupled to current sensing circuit **120**. The current sensing circuit **120** may measure, sense, or detect the current flowing through the regulated power supply node **110**, which may be referred to as an inductor current. The current sensing circuit **120** may also compensate for differences in the temperature coefficients of various FETs, as discussed in more detail below.

[0048] FIG. **4** illustrates a block diagram of an example current sensing circuit **120A**, in accordance with one or more embodiments of the present disclosure. Measuring, detecting, sensing, etc., current may be accomplished using a variety of circuit designs. Current sensing circuit **120A** may be a particular embodiment of current sensing circuit **120** illustrated in FIGS. **1-3**. As illustrated, current sensing circuit **120A** includes current source **411**, current source **413**, FET **421**, FET **423**, FET **425**, inductor **431**, and comparator circuit **440**.

[0049] FET **421**, which may be a particular embodiment of an n-channel MOSFET, is coupled between the current source **411** and the FET **425**. The FET **421** may be another type of FET, such as a p-channel MOSFET in other embodiments. FET **425** is coupled to between FET **421** and a ground node. An inductor **431** (e.g., an inductor circuit or any other appropriate combination of circuits that opposes a change in current) is coupled between the FET **421** and the FET **425**. FET **423**, which may be a particular embodiment of an n-channel MOSFET, is coupled between current source **413** and a ground node. FET **421** and/or FET **423** may be referred to as sense FETs (e.g., a FET that is used to measure/detect current).

[0050] The current source **411** generates a first current (which may be referred to as a demand current) that flows to the ground node via FET **421** (e.g., a sense FET) and FET **425** (e.g., a power FET). An inductor current may also flow through the FET **425**. The current source **413** may generate a second current (which may be referred to as an offset current) that flows to the ground node via FET **412** (e.g., a sense FET).

[0051] In one embodiment, the comparator circuit **440** may include a current comparator and/or any appropriate circuit that compares multiple voltages/currents. As illustrated in FIG. **4**, the comparator circuit **440** also includes a temperature circuit **430**. The temperature circuit **430** may be referred to as an input or an input stage to the comparator circuit **440**. In one embodiment, the temperature circuit **430** may compensate for differences in the temperature coefficients of the FET **421**, FET **423** (e.g., sense FETs), and/or FET **425** (e.g., a power FET). For example, FET **421** and/or FET **423** may have temperature coefficients that are different than the temperature coefficient of FET **425**. The temperature circuit **430** may allow the current sensing circuit **120A** to more accurately sense/measure the current from the current source **411** (e.g., the demand current), even though the FET **421**, FET **423**, and FET **425** may have different temperature coefficients.

[0052] The temperature circuit **430** includes two inputs **441** and **443**. The input **441** is coupled to the current source **411** between the current source **411** and the FET **421**. The input **441** provides a first current from the current source **411** to the temperature circuit **430**. The input **443** is coupled to the current source **413** between the current source **411** and the FET **423**. The input **443** provides a second current from the current source **413** to the temperature circuit **430**.

[0053] In one embodiment, the temperature circuit **430** converts the first current (received from current source **411**) into a first voltage and converts the second current (received from current

source **413**) into a second voltage. The comparator circuit **440** may compare the first voltage and the second voltage as discussed in more detail below. The temperature circuit **430** may also compensate for differences in the temperature coefficients of FET **421**, FET **423**, and/or FET **425**, as discussed in more detail below.

[0054] In one embodiment, the comparator circuit **440** may output, generate, send, etc., a signal when the current flowing through FET **425** (e.g., a power FET) matches a target current. The target current may be determined based on a difference between the current from current source **411** and the current from current source **413**. If the voltage received at input **441** matches the voltage received at input **443**, the comparator circuit may output a signal indicating that the current flowing through FET **425** matches a target current (e.g., a desired, requested, etc., current). The comparator circuit **440** may also generate additional signals to indicate whether the current flowing through FET **425** is greater than or less than the target current.

[0055] In one embodiment, control circuit **101** (illustrated in FIGS. **1** and **3**) may control operation of the power converter circuit **100** based on the signal received from the comparator circuit **440**. For example, the control circuit **101** may halt a charge cycle when the signal (indicating that the first current matches a target or threshold current) is received from the comparator circuit **440**. In another example, the control circuit **101** may initiate/start a charge cycle based on the signal received from the comparator circuit **440**.

[0056] In one embodiment, characteristics of the FET **421** and the FET **423** may be based on the FET **425**. For example, the temperature coefficients of FET **421** and/or the FET **423** may be similar to or within a range/threshold of the temperature coefficient of FET **425**. In a further example, the resistances of FET **421** and/or the FET **423** may be similar to or within a range/threshold of the resistance of FET **425**. In a further example, the size, length, width, etc., of the FET **421** and/or the FET **423** be similar to the size, length, width, etc., of FET **425**.

[0057] Although current source **411** and current source **413** are illustrated in FIG. **4** as being part of the current sensing circuit **120A**, the current source **411** and/or current source **413** may be external to the current sensing circuit **120A** in other embodiments. For example, the current source **411** may be another circuit that is separate from the current sensing circuit **120A** and may be coupled to the current sensing circuit via one or more wires, lines, connections, traces, conduits, etc.

[0058] As illustrated in FIG. **4**, FET **425** may be external to the current sensing circuit **120A**. For example, FET **425** may be FET that is coupled to the current sensing circuit **120A**. In one embodiment, the current sensing circuit **120A** may be part of a power converter circuit **100**. The FET **425** may be external to the power converter circuit **100** but may be coupled to the current sensing circuit **120A** via one or more wires, lines, connections, traces, conduits, etc. In another embodiment, the FET **425** may be a gallium nitride (GaN) FET. The FET **425** may also be referred to as a power FET (e.g., a FET that is used to provide current or power to another circuit/device).

[0059] FIG. **5** illustrates a block diagram of an example current sensing circuit **120B**, in accordance with one or more embodiments of the present disclosure. Current sensing circuit **120B** may be a particular embodiment of current sensing circuit **120** illustrated in FIGS. **1-3**. Measuring, detecting, sensing, etc., current may be accomplished using a variety of circuit designs. A particular embodiment of such a circuit is depicted in FIG. **5**. As illustrated, current sensing circuit **120B** includes current source **511**, current source **513**, FET **521**, FET **523**, FET **525**, inductor **531**, amplifier circuit **540**, capacitor **532**, capacitor **533**, capacitor **535**, capacitor **537**, resistor **561**, and resistor **563**. FETs **521** and **523** may be sense FETs and FET **525** may be a power FET.

[0060] Current source **511** is coupled to resistor **563** and the resistor **563** is coupled to a ground node. Current source **511** is also coupled to a capacitor **537** and the capacitor **537** is coupled to a ground node. The current source **511** includes FET **551** and FET **553**. FET **551** and FET **553** may be particular embodiments of a p-channel MOSFET. The current source **511** is further coupled to an input/source of FET **555**, which may be a particular embodiment of an n-channel MOSFET. The output/drain of FET **555** is coupled to FET **521**. The gate of FET **555** is coupled to an output of the

amplifier circuit **540**.

[0061] FET **521**, which may be a particular embodiment of an n-channel MOSFET, is coupled between FET **555** and the FET **525**. FET **525** is coupled to between FET **521** and a ground node. An inductor **531** (e.g., an inductor circuit or any other appropriate combination of circuits that opposes a change in current) is coupled between the FET **521** and the FET **525** at node LX.

[0062] The current source **511** may generate a first current that flows to the ground node via FET **555**, FET **521**, and FET **525**. In addition, an inductor current may flow through FET **525** (e.g., a power FET) and inductor **531**, as discussed above. The first current (generated by the current source **511**) may be provided to input **543** of the amplifier circuit **540** via FET **555**. In one embodiment, the current sensing circuit **120B** may be referred to as a closed-loop circuit because the output of the current sensing circuit **120B** (e.g., the current that is provided to the FET **525**) is provided back to input **543**.

[0063] In one embodiment, the first current may be an average current. For example, the first current may be an average of a demand current (e.g., a current that is supplied to another circuit/device) over a period of time. The first current generated by the current source **511** may be referred to as an average current or an average sensed current.

[0064] Current source **513** is coupled to FET **523**, a capacitor **532**, and input **541** of the amplifier circuit **540**. Current source **513** may generate a second current (which may be referred to as an offset current) that flows to a ground node via FET **523**. The second current may also be provided to the capacitor **535** and the input **541** of the amplifier circuit **540**.

[0065] The output of amplifier circuit **540** is coupled to the gate of FET **555** and a resistor **561**. The resistor **561** is coupled to a capacitor **535** and the capacitor **535** is coupled to a ground node. As illustrated in FIG. 5, the amplifier circuit **540** includes a temperature circuit **430**. In one embodiment, the amplifier circuit **540** may also include an amplifier (e.g., an operational amplifier or op amp) and/or any appropriate circuit that compares multiple voltages/currents. The temperature circuit **430** may be referred to as an input or an input stage to the amplifier circuit **540**. In one embodiment, the temperature circuit **430** may compensate for differences in the temperature coefficients of the FET **521**, FET **523**, and/or FET **525**. The temperature circuit may allow the current sensing circuit **120B** to sense/measure the current from the current source **511** (e.g., the demand current) more accurately, even though the FET **521**, FET **523**, and/or FET **525** may have different temperature coefficients.

[0066] The temperature circuit **430** includes two inputs. The input **543** is coupled to current source **511** between the FET **555** and the FET **521**. The input **541** provides the first current from current source **511** (which is received via FET **555**) to the temperature circuit **430**. The input **541** is coupled to current source **513**. The input **543** provides a second current (e.g., an offset current from current source **513**) to the temperature circuit **430**.

[0067] In one embodiment, the temperature circuit **430** converts the current received from current source **511** (e.g., an average current) into a first voltage and converts the current received from current source **513** (e.g., an offset current) into a second voltage. The amplifier circuit **540** may compare the first voltage and the second voltage as discussed in more detail below.

[0068] In one embodiment, the amplifier circuit **540** may output, generate, send, etc., a signal and/or voltage that is proportional to the inductor current flowing through FET **525** and inductor **531**. For example, the amplifier circuit **540** and FET **555** may form a negative closed loop. The negative closed loop may regulate the voltage at input **541** and input **543** to the same voltage. In one embodiment, the voltage caused by inductor current flowing through FET **525** and the voltage caused by the current from current source **511** flowing through FET **521** may be equal to the voltage caused by current from current source **513** flowing through FET **523**. Because the current from current source **513** may be fixed and/or known, the current from current source **511** will mirror the inductor current flowing through FET **525**.

[0069] In one embodiment, characteristics of the FET **521** and the FET **523** may be based on the

FET 525. For example, the temperature coefficients of FET 521 and/or the FET 523 may be similar to or within a range/threshold of the temperature coefficient of FET 525. In a further example, the resistances of FET 521 and/or the FET 523 may be similar to or within a range/threshold of the resistance of FET 525. In a further example, the size, length, width, etc., of the FET 521 and/or the FET 523 be similar to the size, length, width, etc. of FET 525.

[0070] Although current source 511 and current source 513 are illustrated in FIG. 5 as being part of the current sensing circuit 120B, current source 511 and current source 513 may be external to the current sensing circuit 120B in other embodiments. For example, the current source 511 may be another circuit that is separate from the current sensing circuit 120B and may be coupled to the current sensing circuit 120B via one or more one or more wires, lines, connections, traces, conduits, etc.

[0071] As illustrated in FIG. 5, FET 525 may be external to the current sensing circuit 120B. For example, FET 525 may be FET that is coupled to the current sensing circuit 120B. In one embodiment, the current sensing circuit 120B may be part of a power converter circuit 100. The FET 525 may be external to the power converter circuit 100 but may be coupled to the current sensing circuit 120B via one or more wires, lines, connections, traces, conduits, etc.

[0072] In various embodiments, FET 521, FET 523, and FET 525 may be particular embodiments of a transmission gate or other suitable combination of FETs, MOSFETs, etc., configured to couple current source 511 and current source 513 to the ground nodes. One or more of capacitor 532, capacitor 533, capacitor 535, and capacitor 537 may be fabricated using a metal-oxide-metal structure, or any other suitable structure available on a semiconductor manufacturing process using to fabricate power converter circuit 100.

[0073] FIG. 6 illustrates a block diagram of an example temperature circuit 430, in accordance with one or more embodiments of the present disclosure. As discussed above, the temperature circuit 430 allows the current sensing circuit 120 (e.g., current sensing circuit 120A and/or current sensing circuit 120B) to compensate for different temperature coefficients of different field effect transistors (FETs). The temperature circuit 430 may be part of a comparator circuit, such as comparator circuit 440 and/or amplifier circuit 540. The temperature circuit 430 may be accomplished using a variety of circuit designs. A particular embodiment of such a circuit is depicted in FIG. 6. As illustrated, temperature circuit 430 includes FET 631, FET 633, FET 635, FET 637, resistor 621, and resistor 623.

[0074] Current source 411 is coupled to a resistor 621 (e.g., a resistor circuit or some other appropriate circuit that provides resistance to the flow of current) and the resistor 621 is coupled to a ground node. The current source 411 is also coupled to a gate of the FET 631. The current source 411 provides a first current (e.g., a demand current) to the resistor 621 and to the gate of the FET 631. Current source 413 is coupled to a resistor 623 and the resistor 623 is coupled to a ground node. The current source 413 is also coupled to a gate of FET 633. The current source 413 provides a second current (e.g., an offset current) to the resistor 623 and to the gate of the FET 633.

[0075] Current source 611 is coupled to the FET 631 and the FET 633. For example, current source 611 is coupled to the inputs/sources of the FET 631 and the FET 633. Current source 611 provides a current (e.g., a bias current) to the FET 631 and the FET 633. Current source 613 is coupled to FET 635 and FET 637. For example, current source 613 is coupled to the inputs/sources of the FET 635 and the FET 637. Current source 613 provides a current (e.g., a bias current) to the FET 635 and the FET 637.

[0076] As discussed above, current source 411 provides a first current (e.g., a demand current) to the resistor 621. The first current may be referred to as $i_{\text{sub.dem}}$. The first current is also provided to the gate of FET 631. The voltage at the gate of FET 631 is represented as V_2 . Also as discussed above, the current source 413 provides a second current (e.g., an offset current) to the resistor 623. The second current may be referred to as $i_{\text{sub.os}}$. The second current is also provided to the gate of FET 633. The voltage at the gate of FET 633 is represented as V_1 . The voltage at inputs 441 and

541 (illustrated in FIGS. **4** and **5**) may be represented as $V_{\text{sub}+}$. The voltage at inputs **443** and **543** (illustrated in FIGS. **4** and **5**) may be represented as $V_{\text{sub}-}$.

[0077] In one embodiment, referring to FIG. **4**, the gate of FET **635** may be coupled to the current source **411** and the gate of FET **637** may be coupled to the current source **413**. In another embodiment, referring to FIG. **5**, the gate of FET **635** may be coupled to the current source **411** and the gate of FET **637** may be coupled to the current source **413**.

[0078] In one embodiment, the comparator circuit that includes the temperature circuit **430** (e.g., comparator circuit **440** and/or comparator circuit **540**) may transmit a signal when the following relationship depicted in Equation 1 is satisfied. The signal may indicate that the first current (e.g., a demand current or an average current) received by the current sensing circuit **120** matches a target current.

$$[00001] V_+ + V_1 = V_- + V_2 \quad (1)$$

[0079] V_+ may be defined as $i_{\text{sub.dem}} * R_{\text{sub.SNS}} - i_{\text{sub.LX}} * R_{\text{sub.powerFET}}$, where $i_{\text{sub.dem}}$ is the demand current, $R_{\text{sub.SNS}}$ is the resistance of the FET **421** and/or FET **423**, $i_{\text{sub.zx}}$ is the inductor current flowing through node LX, and $R_{\text{sub.GAN}}$ is the resistance of the FET **425**. V_1 may be defined as $i_{\text{sub.os}} * R_{\text{sub.res}}$, where $i_{\text{sub.os}}$ is the offset current and $R_{\text{sub.res}}$ is the resistance of the resistors **621** and **623**. $V_{\text{sub}-}$ may be defined $i_{\text{sub.os}} * R_{\text{sub.sNS}}$. V_2 may be defined as $i_{\text{sub.dem}} * R_{\text{sub.res}}$. By substituting in the above defined terms, Equation 2 may be obtained.

$$[00002] (i_{\text{dem}} * R_{\text{SNS}} - i_{\text{LX}} * R_{\text{powerFET}}) + (i_{\text{os}} * R_{\text{res}}) = (i_{\text{os}} * R_{\text{SNS}}) + (i_{\text{dem}} * R_{\text{res}}) \quad (2)$$

[0080] In one embodiment, the temperature circuit **430** may compensate for different temperature coefficients of different FETs by adjusting the resistances of resistor **621** and/or resistor **623** (e.g., by adjusting $R_{\text{sub.res}}$ in Equation 2). For example, resistor **621** and/or resistor **623** may be adjustable resistors (e.g., a resistor or resistor circuit where the resistance can be changed, adjusted, etc.). Adjusting the resistance $R_{\text{sub.res}}$ of the resistor **621** and/or resistor **623** allows the temperature circuit **430** to compensate for differences between the temperature coefficients of the sense FETs (e.g., FET **421**, FET **423**, FET **521**, and/or FET **523**) and the power FETs (e.g., FET **425** and/or FET **525**). For example, the resistance R of the resistor **621** and/or resistor **623** may be adjusted (e.g., increased or decreased) until Equation (1) and/or Equation (2) are satisfied.

[0081] In one embodiment, the temperature circuit **430** may compensate for different temperature coefficients of different FETs by adjusting an offset current (e.g., a second current). For example, the amount of current in the offset current may be adjusted (e.g., increased or decreased). Adjusting the amount of current in the offset current allows the temperature circuit **430** to compensate for differences between the temperature coefficients of the sense FETs (e.g., FET **421**, FET **423**, FET **521**, and/or FET **523**) and the power FETs (e.g., FET **425** and/or FET **525**). For example, the amount of current in the offset current may be adjusted (e.g., increased or decreased) until Equation (1) and/or Equation (2) are satisfied.

[0082] In one embodiment, the temperature circuit **430** may be configured to compensate for different temperature coefficients of different FETs during the manufacturing and/or installation of the power converter circuit **100**. For example, during the manufacturing process of the power converter circuit **100** (which includes the temperature circuit **430**), the temperature circuit **430** may be configured to compensate for different temperature coefficients of different FETs (e.g., resistances and/or amounts of current may be adjusted). In another example, when the power converter circuit **100** is installed into a device (e.g., installed in a computing/electronic device), the temperature circuit **430** may be configured to compensate for different temperature coefficients of different FETs.

[0083] In one embodiment, the temperature circuit **430** may be configured to compensate for different temperature coefficients of different FETs after the power converter circuit **100** has been installed in an electronic/computing device. For example, after the power converter circuit **100** has

been installed in the electronic/computing device and/or during operation of the electronic/computing device, the temperature circuit **430** may be configured to compensate for different temperature coefficients of different FETs.

[0084] Structures such as those shown in FIGS. **1-6** for current sensing may be referred to using functional language. In some embodiments, these structures may be described as including “means for providing a first current from a first current source to a first FET, a temperature circuit, and a power FET,” “means for providing a second current from a second current source to a second FET and the temperature circuit,” “means for compensating for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and a temperature coefficient of the power FET,” “means for determining whether the first current matches a target current,” “means for halting a charge cycle of a power converter circuit in response to determining that the first current matches the target current,” “means for starting a charge cycle of a power converter circuit in response to determining that the first current matches the target current,” “means for adjusting the resistance of one or more resistors,” and “means for adjusting a current from one or more of the first current source and a second current source.”

[0085] The corresponding structure for “means for providing a first current from a first current source to a first FET, a temperature circuit, and a power FET” is current sensing circuit **120** as well as equivalents of this circuit. The corresponding structure for “means for providing a second current from a second current source to a second FET and the temperature circuit” is current sensing circuit **120** as well as equivalents of this circuit. The corresponding structure for “means for compensating for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and a temperature coefficient of the power FET” is temperature circuit **430** as well as equivalents of this circuit. The corresponding structure for “means for determining whether the first current matches a target current” are comparator circuit **440** and/or amplifier circuit **540** and as well as equivalents of these circuits. The corresponding structure for “means for halting a charge cycle of a power converter circuit in response to determining that the first current matches the target current” is control circuit **101** as well as equivalents of this circuit. The corresponding structure for “means for starting a charge cycle of a power converter circuit in response to determining that the first current matches the target current” is control circuit **101** as well as equivalents of this circuit. The corresponding structure for “means for adjusting the resistance of one or more resistors” is current sensing circuit **120** as well as equivalents of this circuit. The corresponding structure for “means for adjusting a current from one or more of the first current source and a second current source” is current sensing circuit **120** as well as equivalents of this circuit.

[0086] FIG. **7** illustrates a flow diagram depicting an embodiment of a method for sensing a current, in accordance with one or more embodiments of the present disclosure. The method, which may be applied to one or more of current sensing circuit **120** and/or power converter circuit **100** as depicted in FIG. **1**, starts at the block **700**.

[0087] The method includes providing a first current from a first current source to a temperature circuit (e.g., temperature circuit **430** illustrated in FIG. **6**) at block **705**. For example, the current sensing circuit may receive the first current (which may be referred to as a demand current, a charge current, etc.) from the first current source. The current sensing circuit may provide the first current to the temperature circuit via one or more wires, lines, traces, connections, nodes, etc.

[0088] The method also includes providing a second current from a second current source to the temperature circuit at block **710**. For example, the current sensing circuit may receive the second current (which may be referred to as an offset current) from the second current source. The current sensing circuit may provide the first current to the temperature circuit via one or more wires, lines, traces, connections, nodes, etc.

[0089] The method further includes compensating for one or more differences between temperature coefficients of the FETs that are included in the current sensing circuit and a power FET at block

715. For example, a power FET may provide the first current to another device/circuit. The FETs of the current sensing circuit may be used to sense the amount of the first current. The temperature circuit may compensate for the differences between temperature coefficients of the FETs of the current sensing circuit and the power FET. In one embodiment, the current sensing circuit and/or temperature circuit may compensate for the differences between temperature coefficients by adjusting the resistance of one or more resistors in the temperature circuit. In another embodiment, the current sensing circuit and/or temperature circuit may compensate for the differences between temperature coefficients by adjusting the amount of second current provided by the second current source (e.g., by adjusting the offset current).

[0090] The method also includes determining whether the first current matches a target current at block **720**. For example, a comparator circuit may determine whether the first current matches the target current using the temperature circuit.

[0091] The method further includes optionally starting or halting (e.g., stopping) a charge cycle based on whether the first current matches the target current at block **725**. For example, if the first current matches the target current, a control circuit may start the charge cycle. In another example, if the first current matches the target current, the control circuit may halt the charge cycle. The method ends at block **799**.

[0092] FIG. **8** illustrates a block diagram of an example computer system **800**, in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. **8**, the computer system **800** includes power management circuit **801**, processor circuit **802**, memory circuit **803**, and input/output circuits **804**, each of which is coupled to power supply signal **805**. In various embodiments, computer system **800** may be a system-on-a-chip (SoC) and/or be configured for use in a desktop computer, server, or in a mobile computing application such as, e.g., a tablet, laptop computer, or wearable computing device.

[0093] Power management circuit **801** includes power converter circuit **100**, which is configured to generate a regulated voltage level on power supply signal **805** in order to provide power to processor circuit **802**, memory circuit **803**, and input/output circuits **804**. Although power management circuit **801** is depicted as including a single power converter circuit, in other embodiments, any suitable number of power converter circuits may be included in power management circuit **801**, each configured to generate a regulated voltage level on a respective one of multiple internal power supply signals included in computer system **800**. In cases where multiple power converter circuits are employed, two or more of the multiple power converter circuits may be connected to a common set of power terminals that connections to power supply signals and ground supply signals of computer system **800**.

[0094] Processor circuit **802** may, in various embodiments, be representative of a general-purpose processor that performs computational operations. For example, processor circuit **802** may be a central processing unit (CPU) such as a microprocessor, a microcontroller, an application-specific integrated circuit (ASIC), or a field-programmable gate array (FPGA).

[0095] Memory circuit **803** may in various embodiments, include any suitable type of memory such as a Dynamic Random-Access Memory (DRAM), a Static Random-Access Memory (SRAM), a Read-Only Memory (ROM), Electrically Erasable Programmable Read-only Memory (EEPROM), or a non-volatile memory, for example. It is noted that although in a single memory circuit is illustrated in FIG. **8**, in other embodiments, any suitable number of memory circuits may be employed.

[0096] Input/output circuits **804** may be configured to coordinate data transfer between computer system **800** and one or more peripheral devices. Such peripheral devices may include, without limitation, storage devices (e.g., magnetic or optical media-based storage devices including hard drives, tape drives, CD drives, DVD drives, etc.), audio processing subsystems, or any other suitable type of peripheral devices. In some embodiments, input/output circuits **804** may be configured to implement a version of Universal Serial Bus (USB) protocol or IEEE 1394

(Firewire®) protocol.

[0097] Input/output circuits **804** may also be configured to coordinate data transfer between computer system **800** and one or more devices (e.g., other computing systems or integrated circuits) coupled to computer system **800** via a network. In one embodiment, input/output circuits **804** may be configured to perform the data processing necessary to implement an Ethernet (IEEE 802.3) networking standard such as Gigabit Ethernet or 10-Gigabit Ethernet, for example, although it is contemplated that any suitable networking standard may be implemented. In some embodiments, input/output circuits **804** may be configured to implement multiple discrete network interface ports.

[0098] Although specific embodiments have been described above, these embodiments are not intended to limit the scope of the present disclosure, even where only a single embodiment is described with respect to a particular feature. Examples of features provided in the disclosure are intended to be illustrative rather than restrictive unless stated otherwise. The above description is intended to cover such alternatives, modifications, and equivalents as would be apparent to a person skilled in the art having the benefit of this disclosure.

[0099] The scope of the present disclosure includes any feature or combination of features disclosed herein (either explicitly or implicitly), or any generalization thereof, whether or not it mitigates any or all of the problems addressed herein. Accordingly, new claims may be formulated during prosecution of this application (or an application claiming priority thereto) to any such combination of features. In particular, with reference to the appended claims, features from dependent claims may be combined with those of the independent claims and features from respective independent claims may be combined in any appropriate manner and not merely in the specific combinations enumerated in the appended claims.

Claims

1. An apparatus, comprising: a first field-effect transistor (FET) coupled to a first current source, an inductor, and a power FET; a second FET coupled to a second current source; and a temperature circuit coupled to the first FET and the second FET, the temperature circuit configured to compensate for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and the power FET.
2. The apparatus of claim 1, wherein the temperature circuit comprises: a third FET and a fourth FET coupled in parallel; a fifth FET and a sixth FET coupled in parallel; a first resistor coupled to the first current source; and a second resistor coupled to the second current source.
3. The apparatus of claim 2, wherein a gate of the third FET is coupled to the first current source, a gate of the fourth FET is coupled to the second current source, a gate of the fifth FET is coupled to the first current source; and a gate of the sixth FET is coupled to the second current source.
4. The apparatus of claim 1, wherein the inductor is coupled between the first FET and the power FET.
5. The apparatus of claim 1, further comprising: a comparator circuit, wherein the comparator circuit comprises the temperature circuit and wherein the temperature circuit comprises an input for the comparator circuit.
6. The apparatus of claim 5, wherein the comparator circuit is configured to output a signal indicating when a first current from the first current source matches a target current.
7. The apparatus of claim 5, wherein the comparator circuit is coupled to a control FET and wherein the control FET is configured to start or halt a charge cycle based on a signal generated by the comparator circuit.
8. The apparatus of claim 1, wherein the temperature circuit further comprises one or more resistors and wherein to compensate for the one or more differences between the one or more temperature coefficients, the temperature circuit is configured to adjust one or more resistances of the one or

more resistors.

9. The apparatus of claim 1, wherein to compensate for the one or more differences between the one or more temperature coefficients, the temperature circuit is configured to adjust a current from one or more of the first current source and the second current source.

10. The apparatus of claim 1, wherein the power FET is external to the apparatus.

11. A method, comprising: providing a first current from a first current source to a first FET, a temperature circuit, and a power FET; providing a second current from a second current source to a second FET and the temperature circuit; and compensating for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and a temperature coefficient of the power FET.

12. The method of claim 11, further comprising: determining whether the first current matches a target current.

13. The method of claim 12, further comprising: in response to determining that the first current matches the target current, halting a charge cycle of a power converter circuit.

14. The method of claim 12, further comprising: in response to determining that the first current matches the target current, starting a charge cycle of a power converter circuit.

15. The method of claim 11, wherein compensating for the one or more differences between the one or more temperature coefficients comprises adjusting one or more resistances of one or more resistors.

16. The method of claim 11, wherein compensating for the one or more differences between the one or more temperature coefficients comprises adjusting a current from one or more of the first current source and the second current source.

17. An apparatus, comprising: a voltage regulator circuit comprising a switch node coupled to a regulated power supply node via an inductor, wherein the voltage regulator circuit is configured to source a charge current to the switch node during a charge cycle; a current sensing circuit comprising: a first field-effect transistor (FET) coupled to a first current source, the inductor, and a power FET; a second FET coupled to a second current source; and a comparator circuit coupled to the first FET and the second FET, the comparator circuit configured to compensate for one or more differences between one or more temperature coefficients of one or more of the first FET, the second FET, and the power FET and configured to determine whether the charge current from the first current source matches a target current; and a control circuit configured to, in response to determining that the charge current matches the target current, start the charge cycle or stop the charge cycle.

18. The apparatus of claim 17, wherein the comparator circuit further comprises one or more resistors and wherein to compensate for the one or more differences between the one or more temperature coefficients, the comparator circuit is configured to adjust one or more resistances of the one or more resistors.

19. The apparatus of claim 17, wherein to compensate for the one or more differences between the one or more temperature coefficients, the comparator circuit is configured to adjust a current from one or more of the first current source and the second current source.

20. The apparatus of claim 17, wherein the inductor is coupled between the first FET and the power FET.
